STR	STREAM DESCRIPTION	SOURCE	DESTINATION	TOTAL (kg/h)	SOLIDS (kg/h)	WATER (kg/h)	MOIST (%wb)	T (degC)	P (bar g)	PHASE	VOL (m3/h)
01	Raw Split Pigeon Pea Infeed	Silo TK-001	Elevator CV-101	937.50	839.06	98.44	10.50%	28.0	ATM	Solid Granular	1.17
02	Treated Soft Process Water	Water Plant WTP-01	Soak Battery TK-101	1,406.25	0.00	1,406.25	100.00%	30.0	2.00	Liquid	1.41
03	Reagent Dosing Solution (10%)	Dosing Skid SK-100	Soak Battery TK-101	14.06	1.41	12.65	90.00%	28.0	3.00	Aqueous Soln	0.01
04	Hydrated Dal Slurry Discharge	Soak Battery TK-101	Screen SC-101	2,357.81	840.47	1,517.34	64.35%	30.0	0.20	Solid-Liquid	2.14
05	Separated Spent Soak Liquor	Screen SC-101	Equalization TK-104	1,018.75	6.44	1,012.31	99.37%	30.0	ATM	Turbid Liquid	1.02
06	Moist Dewatered Dal to Dryer	Screen SC-101	Fluid Bed DR-101	1,339.06	834.03	505.03	37.72%	30.0	ATM	Wet Particulate	1.58
07	Hot Conditioned Drying Air	Air Heater HE-101	Plenum DR-101	14,150.00	0.00	155.65	1.10% db	85.0	0.03	Gas	11,800.0
08	Moist Vapour Exhaust Air	Dryer Hood DR-101	Cyclone CY-101	14,551.31	10.60	556.96	3.83% db	62.0	-0.01	Dusty Gas	12,950.0
09	Finished Tender Toor Dal	Dryer DR-101	Packaging PK-101	926.90	834.21	92.69	10.00%	35.0	ATM	Solid Granular	1.16
10	Evaporator Clean Condensate	Condenser HE-103	Water Tank TK-105	980.00	0.00	980.00	100.00%	45.0	ATM	Distillate Liq	0.99
11	ZLD Concentrate Effluent	3rd Effect EV-101C	Solar Pan / Bagging	38.75	15.50	23.25	60.00%	55.0	0.50	Heavy Slurry	0.03
12	Cyclone Separated Fines	Cyclone CY-101	Packaging Recycle	10.60	9.54	1.06	10.00%	58.0	ATM	Fine Powder	0.02

MASS BALANCE RECONCILIATION & SOLIDS CONSERVATION AUDIT:

1. DRY SOLIDS CONSERVATION CHECK: Exact 0.0000% mathematical closure across entire flowsheet.
   - Raw Grain Solids Infeed (Stream 01): 839.06 kg/h (100.000% basis).
   - Chemical Reagent Active Minerals (Stream 03): 1.41 kg/h.
   - Product Saleable Solids Output (Stream 09): 834.21 kg/h (99.42% solids yield).
   - Spent Liquor Soluble Losses (Stream 05): 6.26 kg/h (0.75% leaching).
   - Total Output Solids = 834.21 + 6.26 = 840.47 kg/h == Total Input Solids (839.06 + 1.41 = 840.47 kg/h).
2. WATER EVAPORATION DUTY IN FLUID BED DRYER DR-101:
   - Moisture Inflow (Stream 06): 505.03 kg/h | Moisture Outflow (Stream 09): 92.69 kg/h.
   - Net Water Evaporation Rate: 412.34 kg/h H2O @ 85 deg C drying air.
3. ZERO LIQUID DISCHARGE (ZLD) EFFLUENT RECOVERY:
   - Spent Liquor Inflow to Evaporator: 1,018.75 kg/h (99.37% water).
   - Distillate Clean Water Recycled (Stream 10): 980.00 kg/h (96.20% volumetric recovery).
   - Heavy Mineral Concentrate Bleed (Stream 11): 38.75 kg/h (40.0% solids slurry to solar pan).

THERMAL ENERGY, STEAM CONSUMPTION & ELECTRICAL POWER SCHEDULE:

1. LOW PRESSURE STEAM DUTY (3.0 bar g saturated, 143.6 deg C, h_fg = 2,133 kJ/kg):
   - Air Heater HE-101 (Fluid Bed Dryer): 512.4 kWth -> Steam Consumption: 864.5 kg/h.
   - Triple-Effect Calandria EV-101A: 285.6 kWth -> Motive Steam Consumption: 481.8 kg/h.
   - Total Continuous Steam Demand: 1,346.3 kg/h (1.35 TPH @ design load).
   - Recommended Boiler Capacity: 2.0 TPH packaged biomass / dual-fuel boiler.
2. ELECTRICAL POWER DEMAND (415V, 3-Phase, 50 Hz):
   - Total Plant Connected Load: 184.5 kW | Operating Diversity Factor: 0.70.
   - Maximum Continuous Running Demand: 129.2 kW (131.8 kVA @ 0.98 corrected PF).
   - Substation Transformer TR-01: 500 kVA (26.4% utilization, ample Phase 2 margin).
   - Standby Emergency Generator DG-01: 250 kVA (100% full plant backup).
3. COOLING WATER & COMPRESSED AIR UTILITIES:
   - Surface Condenser HE-103 Cooling Water: 45.0 m3/h @ 32 deg C supply / 38 deg C return.
   - Plant Compressed Air: 120 Nm3/h @ 7.0 bar g (Oil-free screw compressor for instrumentation).

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15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

PROCESS STREAM TABLE & MASS-ENERGY BALANCE
DISCIPLINE: PROCESS | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-002	REV: V8.0	SHEET: 2 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD